Tony Willis

CNC Programmer | CAD/CAM Applications Specialist | CNC Trainer

Upminster, London · email and phone number on request · tam-cnc.com bonesforge.co.uk · Full UK driving licence (car and motorcycle)

PROFILE

CNC programmer and workshop technician with over 20 years of hands-on precision machining experience in UK university engineering departments, including 13 years as Senior Research Workshop Technician at Imperial College London. Fluent in G-code (written by hand and generated through CAM) and experienced in CAMWorks on SolidWorks, Fusion 360 and FeatureCAM. Creator of TAM-CNC, a browser-based virtual CNC milling machine for teaching machine operation, and of a companion Android app that generates G-code programs from simple inputs. Experienced in training and mentoring students and staff. Looking for a remote or hybrid applications engineering, technical support or CNC training role.

KEY SKILLS

CNC machines: Haas TL1, TM2, VF2 (4 axis) and ST20Y (4 axis); Sodick Wire EDM AQ400L and VL400Q; XYZ mills and lathes

Programming: Handwritten G-code, CAM-generated programs, toolpath optimisation, program proving; custom VQC programs on Haas

CAD/CAM: SolidWorks with CAMWorks, Fusion 360, Creo Pro Engineer, FeatureCAM; reverse engineering and 3D modelling

Software and scripting: Python (G-code generation), HTML/CSS/JavaScript, Git/GitHub

Training and support: One-to-one and group instruction, technical mentoring, demonstrations, writing learning material

Also: Microsoft Azure Fundamentals (AZ-900); Microsoft Entra single sign-on concepts

EXPERIENCE

Senior Research Workshop Technician Feb 2013 – Present

Department of Mechanical Engineering, Imperial College London

- Program and operate a full suite of CNC milling, turning and EDM equipment to produce one-off, tight-tolerance parts for research projects, including turbine test rigs, material test samples and bespoke medical devices (cranial implants for Eden 2020).
- Write and debug G-code by hand and through CAM (CAMWorks on SolidWorks, Fusion 360), optimising toolpaths and proving programs for safe operation before cutting.
- Wrote custom VQC programs for the Haas ST20Y that turn a short list of operator inputs into a complete part program.
- Reverse engineer parts and design components in SolidWorks, turning researchers' requirements into manufacturable parts.
- Train and mentor students and staff on CNC machines, lathes and workshop equipment, from basic turning to micro hardness testing.
- Maintain heritage engineering projects, including the college's mascot vehicle and original tower clock.

Engineer / Technician 2003 – 2013

School of Engineering and Materials Science, Queen Mary University of London

- Fabricated components on manual and CNC machinery and supported student design and prototyping.
- Gave demonstrations to student groups and visiting cohorts, and led safe use of measurement tools.
- Maintained mechanical laboratories and installed lab equipment across engineering and biology departments; managed equipment ordering, inventory and safety compliance.

Machine Operator 1999 – 2003

Getty Images, London

- Ran automated framing systems, logged performance, managed maintenance cycles and ordered critical spares; supported digital imaging workflows.

PROJECTS

TAM-CNC: virtual CNC training platform In development

tam-cnc.com

- Designed and built a browser-based virtual CNC milling machine that teaches the basics of machine operation, CAD/CAM and G-code without needing access to a real machine.
- Being developed for licensing to schools and colleges, with sign-in through the institution's own Microsoft Entra accounts.

TAM-CNC Android app

tam.bonesforge.co.uk

- Wrote the Python program behind an Android app that generates basic G-code programs from a simple input list.

EDUCATION AND CERTIFICATIONS

- HNC in Mechanical Engineering
- NVQ in Mechanical Engineering
- A Level Physics; GCSEs
- Microsoft Azure Fundamentals (AZ-900)

INTERESTS

Motorcycling, home workshop, electronics, (hobby) blacksmithing and historical re-enactment builds.

References available on request.